Introduction to Machine Learning

Page 1: What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Common applications include email spam detection, product recommendations, image recognition, speech recognition, and fraud detection.

Page 1 Key Point

Machine learning systems use data and algorithms to improve their performance on a task.

Page 2: Types of Machine Learning

Supervised learning uses labeled data. Each training example contains input data and a known correct output. Examples include predicting house prices and classifying emails as spam or not spam.

Unsupervised learning uses data without labeled answers. It finds hidden patterns or groups. Examples include customer segmentation and grouping similar news articles.

Reinforcement learning trains an agent through rewards and penalties as it interacts with an environment. Examples include game-playing agents and robot navigation.

Page 2 Key Point

The three common learning types are supervised, unsupervised, and reinforcement learning.

Page 3: The Machine Learning Workflow

A typical machine learning project begins by collecting data. The data is then cleaned and prepared. Next, the dataset is divided into training and testing sets. A model is trained on the training data and evaluated on the testing data.

If the results are not satisfactory, the developer may improve features, tune model settings, or collect more data. The final model can then be deployed and monitored.

Page 3 Key Point

A reliable workflow includes data collection, preprocessing, training, evaluation, deployment, and monitoring.